

WEEK : 3 Seller and Inventory Management System

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Scenario:

The Art Work Shop wants to maintain seller and inventory information. The system stores seller details and tracks product stock quantity, stock status, and the date of the latest inventory update.

1.Creating Seller and Inventory tables

```
Connected.
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(7), Seller_Name VARCHAR(25),Email VARCHAR(30), Phone number(10), Address VARCHAR(45));
Table created.

SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(7),
 2 Product_ID VARCHAR(7), Seller_ID VARCHAR(7),
 3 Stock_Quantity number,Stock_status VARCHAR(11),
 4 Last_Updated DATE);
Table created.
```

2.Establish Relationship using Primary Key

The primary key is used to uniquely identify each record in the Seller and Inventory tables. Seller_ID is the primary key of the Seller table, and Inventory_ID is the primary key of the Inventory table.

Seller → Product = One-to-Many (1 : M)

One seller can sell many products, while each product is associated with one seller.

Product → Inventory = One-to-Many (1 : M)

One product can have many inventory records, while each inventory record is associated with one product.

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);
Table altered.
```

3. Foreign key for inventory table:

Inventory table is connected with Product and Seller tables using foreign keys.

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINTS FK_prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);  
  
Table altered.
```

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Seller FOREIGN KEY (Seller_ID) REFERENCES Seller (Seller_ID);  
  
Table altered.  
  
SQL>
```

4. Constraints to make Unique Email ID:

The UNIQUE constraint is added to the Email column in the Seller table to ensure that no two sellers can have the same email ID.

```
connected.  
SQL> ALTER TABLE Seller ADD CONSTRAINT Un_mail UNIQUE(Email);  
  
Table altered.
```

To make Email NOT NULL:

The NOT NULL constraint ensures that every seller record must have an email ID.

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;  
  
Table altered.
```

Add Constraint to Check Stock Quantity is Positive:

The CHECK constraint ensures that the stock quantity in the Inventory table is always greater than zero.

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT CK_Quan CHECK(Stock_Quantity>=0);  
  
Table altered.
```

5. Inserting Values into Seller Table

```
SQL> INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
2 ;
Enter value for sellid: S00001
Enter value for sellername: Malar Enterprises
Enter value for email: mdmalarenterpri@gmail.com
Enter value for phone: 9876543210
Enter value for address: 33,sellai Nagar,Chennai
old 1: INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00001','Malar Enterprises','mdmalarenterpri@gmail.com',9876543210,'33,sellai Nagar,Chennai')

1 row created.

SQL> /
Enter value for sellid: S00002
Enter value for sellername: Vasanth & CO
Enter value for email: mdvasanthandco@gmail.com
Enter value for phone: 8765457890
Enter value for address: 45,dubai cross street,kanyakumari
old 1: INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00002','Vasanth & CO ','mdvasanthandco@gmail.com',8765457890,'45,dubai cross street,kanyakumari')

1 row created.

SQL> /
Enter value for sellid: S00003
Enter value for sellername: Sathya Ricemill
Enter value for email: mdsathyarice@gmail.com
Enter value for phone: 7654321890
Enter value for address: 12, vjsiddhu street,kumbakonam
old 1: INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00003','Sathya Ricemill','mdsathyarice@gmail.com',7654321890,'12, vjsiddhu street,kumbakonam')

1 row created.

SQL> /
Enter value for sellid: S00004
Enter value for sellername: RK Hospital
Enter value for email: mdmanasaceo@gmail.com
Enter value for phone: 7860984567
Enter value for address: 65, reena cross street,Bangalore
```

```
SQL> /
Enter value for sellid: S00005
Enter value for sellername: Poorvika enterprises
Enter value for email: mdpoorvika
Enter value for phone: 9876789098
Enter value for address: 23,gandhi nagar,medavakkam
old 1: INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00005','Poorvika enterprises','mdpoorvika',9876789098,'23,gandhi nagar,medavakkam')

1 row created.

SQL> /
Enter value for sellid: S00006
Enter value for sellername: FROG Enterprises
Enter value for email: mdfrogenterp@gmail.com
Enter value for phone: 8907890657
Enter value for address: 5,jakuvar street,rajasthan
old 1: INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00006','FROG Enterprises','mdfrogenterp@gmail.com',8907890657,'5,jakuvar street,rajasthan')

1 row created.
```

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
S00001	Malar Enterprises 33, sellai Nagar, Chennai	mdmalarenterpri@gmail.com	9876543210
S00002	Vasanth & CO 45, dubai cross street, kanyakumari	mdvasanthandco@gmail.com	8765457890
S00003	Sathya Ricemill 12, vjsiddhu street, kumbakonam	mdsathyarice@gmail.com	7654321890
S00004	RK Hospital 65, reena cross street, Bangalore	mdmanasaceo@gmail.com	7860984567
S00005	Poorvika enterprises 23, gandhi nagar, medavakkam	mdpoorvika	9876789098
S00006	FROG Enterprises 5, jakuvar street, rajasthan	mdfrogenterp@gmail.com	8907890657

6 rows selected.

6. Inserting Values to Inventory table

```
SQL> INSERT INTO Inventory
  2 VALUES ('Inv0001', 'P1001', 'S00001', 4, 'Available', TO_DATE('2026-09-12', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES ('Inv0002', 'P1002', 'S00002', 3, 'Available', TO_DATE('2026-09-13', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES ('Inv0003', 'P1003', 'S00003', 4, 'Unavailable', TO_DATE('2026-09-15', 'YYYY-MM-DD'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES ('Inv0004', 'P1004', 'S00004', 3, 'Available', TO_DATE('2026-09-14', 'YYYY-MM-DD'));

1 row created.
```

```
SQL> SELECT * FROM INVENTORY;
```

INVENTO	PRODUCT	SELLER_	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
Inv0001	P1001	S00001	4	Available	12-SEP-26
Inv0002	P1002	S00002	3	Available	13-SEP-26
Inv0003	P1003	S00003	4	Unavailable	15-SEP-26
Inv0004	P1004	S00004	3	Available	14-SEP-26

REPORT 1 : Seller wise Product Report

This report displays the number of products handled by each seller using the Seller, Inventory, and Product tables.

Purpose: To identify how many different products are associated with each seller.

```
SQL> SELECT s.Seller_Name,
2         COUNT(i.Product_ID) AS Number_of_Products,
3         SUM(i.Stock_Quantity) AS Available_Stock
4 FROM Seller s
5 JOIN Inventory i ON s.Seller_ID = i.Seller_ID
6 WHERE i.Stock_Status = 'Available'
7 GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
Malar Enterprises	1	4
RK Hospital	1	3
Vasanth & CO	1	3

Report 2: Stock Availability Report

This report shows the product name, available stock quantity, and current stock status.

Purpose: To check the current stock quantity and availability status of each product.

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status FROM Product p JOIN Inventory i ON p.Product_ID = i.Product_ID;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATU
Laptops	4	Available
Croptops and shirts	3	Available
Fish tank	4	Unavailable

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATU
Television	3	Available

Report 3: Inventory Analysis

This report analyzes the inventory to find the total available products, products that are out of stock, the product with the highest stock quantity, and the average inventory quantity.

TO FIND :

➤ Total Available products:

```
SQL> SELECT COUNT(*) AS Total_Available_Products FROM Inventory WHERE Stock_Status = 'Available';
```

TOTAL_AVAILABLE_PRODUCTS
3

➤ Product Out of Stock:

```
SQL> SELECT COUNT(*) AS OutOfStock_Products FROM Inventory WHERE Stock_Status = 'Out of Stock';
```

OUTOFSTOCK_PRODUCTS
0

➤ Highest Stock Product:

```
SQL> SELECT p.Product_Name, i.Stock_Quantity FROM Product P JOIN Inventory i ON p.Product_ID = i.Product_ID WHERE I.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);
```

PRODUCT_NAME	STOCK_QUANTITY
Laptops	4
Fish tank	4

➤ Average Inventory Quantity:

```
SQL> SELECT AVG(Stock_Quantity)AS Average_Inventory FROM Inventory;
```

AVERAGE_INVENTORY
3.5